

HAMZA JEDDAD

Generative AI & Agentic Platforms Expert

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📍 Paris, France

PROFESSIONAL SUMMARY

AI/ML Engineer with 6+ years of experience building **production Generative AI systems, enterprise chatbots, and intelligent automation workflows**. Deep expertise in **LLM orchestration, RAG, agentic patterns (tool-calling, multi-agent), and human-in-the-loop** validation. Proven track record bridging the gap between platform capabilities and business value — translating complex technical architectures into clear, actionable knowledge for cross-functional stakeholders and end-users. Experienced in **agile delivery, PoC prototyping, and full remote collaboration**.

WORK EXPERIENCE

Senior AI Engineer

Leboncoin

📅 Nov 2025 – Feb 2026

📍 Paris, France

- Contributed to **Markus**, Leboncoin's customer support chatbot serving millions of users, designing **conversational AI flows** with LLM-based dialogue management, intent classification, entity extraction, and context-aware response generation.
- Built **RAG pipelines** grounding chatbot responses in knowledge bases, FAQs, and policy documents — reducing hallucinations and improving answer accuracy.
- Designed **autonomous triage and escalation logic** using agentic patterns to minimize manual intervention while maintaining quality, intelligently routing complex cases.
- Implemented **guardrails and safety mechanisms** for edge cases and inappropriate queries, with seamless escalation to human agents (human-in-the-loop).
- Architected integration patterns connecting the chatbot to backend systems, knowledge bases, and ticketing workflows — ensuring **GDPR compliance** and data sovereignty.
- Created **technical documentation and walkthroughs** of chatbot architecture, flows, and integration patterns for cross-functional teams (product, UX, support).
- Monitored performance metrics (resolution rate, satisfaction, escalation rate) and drove continuous improvements based on production data and user feedback.

Stack: Python, LangChain, LangGraph, OpenAI/Anthropic APIs, RAG, FastAPI, Docker, AWS/GCP

Senior AI Engineer

Mérieux NutriSciences

📅 Oct 2025 – Feb 2026

📍 Full Remote

- Built a matching engine combining **rules + fuzzy matching** with **LLM-assisted mapping (Gemini with tool/function calling)** for candidate generation and validation, applying confidence scores and thresholds.
- Implemented **human-in-the-loop review**: results routed to reviewers for approval/correction; captured feedback for continuous improvement and auditability.
- Engineered ingestion, validation, and export tools supporting repeatable batch jobs and future real-time endpoints.
- Documented system architecture and created **reusable how-to guides** for onboarding and knowledge transfer.

Stack: Python, Gemini (function calling); AWS

Senior AI/ML Engineer

Interencheres

📅 Nov 2024 – Oct 2025

📍 Paris, France

- Built production AI systems for catalog enrichment and pricing across **3M items/year**. Collaborated daily with business experts (métier) and backend engineers; added **human-in-the-loop** validation to all LLM services.
- AI Classification (hierarchical, multi-label)**
 - Designed a tree-based taxonomy (domain → subdomain → category → subcategory), expanding from **63 → 108** categories; supported **up to 3 labels** per domain.
 - Hierarchical prompting** with **LangChain/LangGraph**; each node returns **Pydantic-validated JSON** (label, confidence, justification). Guardrails: schema checks, strict node lists, and context minimization to reduce hallucinations.
- Built a Flask annotation platform for sampling-based review by seller/period/region; tracked experiments and metrics in MLflow. Added seller post-prod validation to measure live accuracy. Created **how-to guides and documentation** covering taxonomy design, prompting strategies, and annotation workflows.

- Benchmarked multiple providers (OpenAI mini, Bedrock Titan/Llama, Gemini) for latency/cost/quality; selected **OpenAI mini** as best trade-off.
- **Outcome: 95% accuracy** on validation; powering category filters on site.
- **Category-Aware Characteristics Extraction**
 - From raw descriptions/images, extracted **category-dependent attributes** (e.g., cars: brand/model/mileage; jewelry: material/weight) to unlock **filters, dashboards, similar-item retrieval**.
 - Prompt-based extraction with strict JSON schema, multilingual models, unit normalization, and **ReAct-style reasoning** requiring evidence/justification per field to limit hallucinations.
 - Same HITL workflow via annotation UI; deployed as FastAPI on AWS with Docker; shared observability via MLflow.
- **Image-Driven Item Valuation (advanced prototype)**
 - Pipeline: DINOv2-Large embeddings → Qdrant vector search (visual comps) → LLM image description → classification → Elasticsearch more_like_this (+ category filter) → LLM similarity judge & query refinement → LLM price estimation from comparables.
 - Achieved **30% average error** on historical hammer prices. Status: experimental; integrating feedback loops and cost/latency controls.
- Infra: FastAPI, Docker, AWS EC2/SageMaker, asyncio; tooling: LangChain, LangGraph, MLflow, Pydantic, Qdrant, Elasticsearch; LLMs: OpenAI mini, Bedrock (Titan/Llama), Gemini.

Stack: Python, LangChain, LangGraph, MLflow, Pydantic, FastAPI, Flask, Docker, AWS EC2/SageMaker, Qdrant, Elasticsearch, OpenAI/Bedrock/Gemini, asyncio

Senior ML Engineer

AntVoice

📅 Nov 2022 – Oct 2024

📍 Paris, France

- Contributed to AntVoice's DSP managing ad campaigns on the open web (RTB). Operated at **1B+ queries/day** with strict latency and budget constraints.
- **RTB Prediction Models (CTR/CVR/Viewability/Qualification)**
 - Built models for post-impression and post-view clicks, conversions, view rate, and post-click qualification.
 - Designed the bid pricing function ($price = f(p_{click}, p_{view}, p_{conv}, \dots)$) integrating market dynamics, win-rate, and margin constraints under fixed monthly budgets.
 - Led feature engineering & audience segmentation (retargeting vs. acquisition). **Result: +20% CTR** (1.0 → 1.2%).
- **A/B Testing & Experimentation**
 - User-level randomized traffic split (A/B/C) to challenge new models fairly against baseline; near real-time monitoring in Looker Studio.
- **MLOps & Monitoring**
 - Orchestrated pipelines with Airflow; containerized deployments to Vertex AI. Monitoring for feature drift, unseen values, performance degradation; threshold-based retraining; human-in-the-loop approval for production.
- **Collaboration & Reporting**
 - Weekly sessions with account managers; monthly product updates to cross-functional stakeholders — translating technical ML metrics into business impact.

Stack: Python, Scikit-learn, CatBoost, ONNX, Airflow, Vertex AI, BigQuery, Looker Studio, Docker, GCP

Data Scientist

Direct Assurance (AXA Group)

📅 Jan 2020 – Oct 2022

📍 Paris, France

- Worked across two departments: **Claims** (automation) and **Actuarial** (commercial premium optimization).
- **Claims Automation**
 - Built **OCR+NLP extraction** for diverse documents (invoices, records, forms) with rule-based validation.
 - Implemented RPA (Selenium) and overnight batch jobs to automate data entry — replacing fully manual processes and reducing lead time and errors.
- **Commercial Premium Optimization (Actuarial)**
 - Built conversion models (LogReg, CatBoost) capturing price elasticity; combined with pure premium and renewal models to forecast **7-year CLV**.
 - Two-step optimization: (1) simulate price scenarios under constraints; (2) train meta-model mapping features to optimal margin.
 - Delivered **Dash/Plotly dashboards** and **Databricks notebooks (PySpark)** for A/B testing. Cadence: new releases every 2 months with typical +0.5pt gains per iteration.
- Collaboration: actuaries, data engineers, and business teams; field discovery with métier for claims automation.

Stack: Python, SQL, Azure, Databricks, PySpark, CatBoost, Scikit-learn, Dash, Plotly, Selenium

NLP Research Intern

AXA France

📅 May 2019 – Oct 2019

📍 Paris, France

- Fine-tuned **BERT-based NER models** for unstructured insurance documents (beneficiary, claim type, coverage) using transfer learning.
- Improved extraction quality through domain-specific tokenization and careful hyperparameter tuning.

Tools: *BERT, Transformers, PyTorch, TensorFlow*

Developer & Data Science Intern

Paris Digital Lab

📅 Feb 2017 – Jul 2017

📍 Paris, France

- Built recommendation systems and **chatbots** for clients (AccorHotels, Foncia) to automate customer support and engagement.
- Managed delivery from requirements to working prototypes — demonstrating early **PoC-to-production** capability.

Tools: *Python, NLP, Machine Learning, Pandas, Scikit-learn*

EDUCATION

Master of Science: Artificial Intelligence

CentraleSupélec (École Centrale Paris)

📅 Sept 2018 – Dec 2019

📍 Paris, France

Master of Engineering: Applied Mathematics & Computer Science

CentraleSupélec (École Centrale Paris)

📅 Sept 2015 – Dec 2019

📍 Paris, France

SKILLS

- **GenAI & LLMs:** Conversational AI, LLM orchestration, LangChain, LangGraph, RAG, ReAct, prompt engineering, fine-tuning, multimodal flows, guardrails, human-in-the-loop.
- **Agentic AI:** Autonomous agents, tool-calling, multi-agent orchestration, hierarchical prompting, autonomous triage and escalation.
- **Enterprise Integration:** Knowledge base integration, ticketing systems, GDPR compliance, data sovereignty, IT automation.
- **Technical Communication:** Technical documentation, how-to guides, PoC walkthroughs, cross-functional stakeholder presentations, business-to-technical translation.
- **Machine Learning:** Scikit-learn, CatBoost, XGBoost, PyTorch, TensorFlow, ONNX, classification, regression, feature engineering.
- **Data Engineering:** BigQuery, PostgreSQL, MongoDB, Elasticsearch, Qdrant, Airflow, Spark, vector databases.
- **MLOps:** Vertex AI, SageMaker, Docker, CI/CD, observability, monitoring, drift detection, experiment tracking (MLflow).
- **Software Engineering:** Python, FastAPI, asyncio, Flask, Dash, Git, testing, code reviews, Pydantic validation.
- **Cloud:** AWS, GCP, Azure.
- **Visualization & Reporting:** Dash, Plotly, Looker Studio.

LANGUAGES

English: Professional proficiency (near-native)

French: Bilingual

Arabic: Native